

Key features



1 ∞

2

a As x increases, the function increases at a slower and slower rate.

b $(0,0)$

c No

d No

e ∞

3

a

x	0	1	2	3	4	5	9
y	0	-1	-1.41	-1.73	-2	-2.24	-3

b No

c Decreasing function

d The function decreases at a slower and slower rate.

4

a

x	-5	-4	-3	-2	-1	0
y	2.24	2	1.73	1.41	1	0

b Decreasing

5

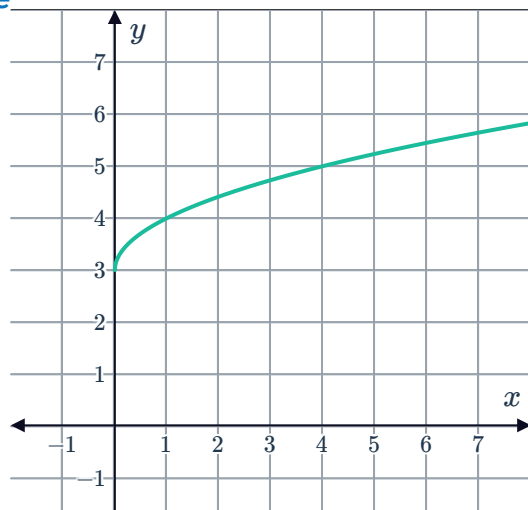
a No

b ∞

c $y = 3$

d 0

e





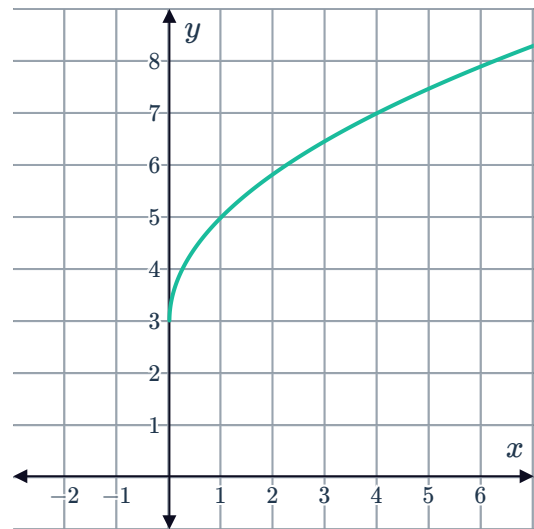
6

a Increasing

c $(0, 3)$

b More steep

d



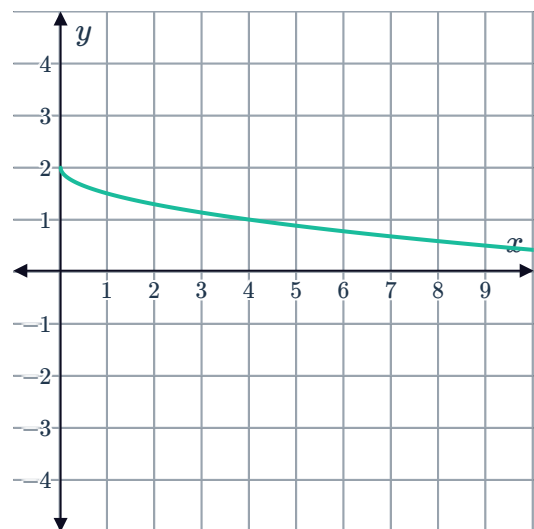
7

a Decreasing

c $(0, 2)$

b Less steep

d



Domain and range

8

a $x \leq 0$

b $y \geq 0$

9

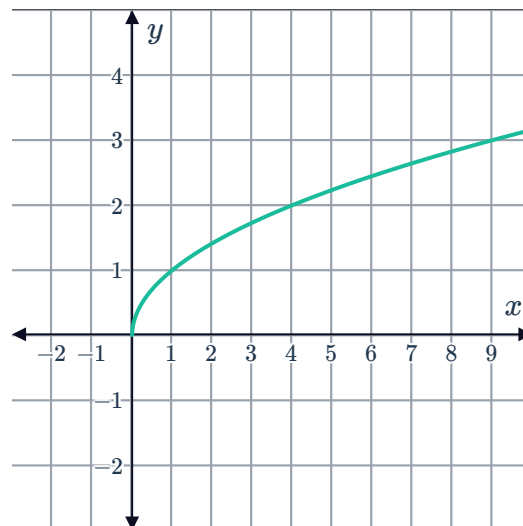


a

x	0	1	2	3	4	5	9
y	0	1	1.41	1.73	2	2.24	3

b $x \geq 0$ c $y \geq 0$ d ∞

e



10

a $x \geq 0$ b $y \leq 0$ 11 $x \geq 0$ and $y \geq -2$

12

a $x \geq 5$ b $y \geq 0$

c Yes

13

a 6

b $x \leq 0$ c $y \geq 6$

14

a No

b No

c Yes

d No

e Yes

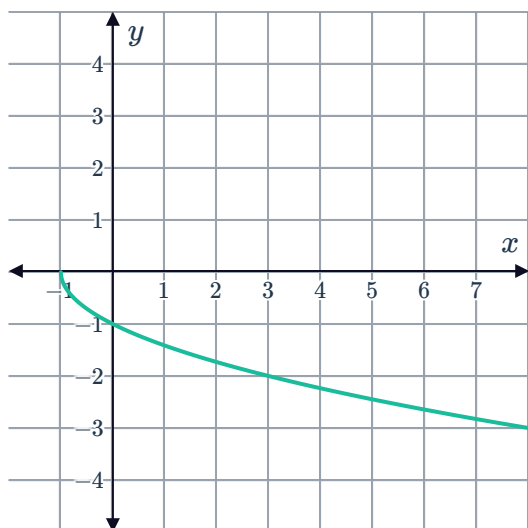
f No

15



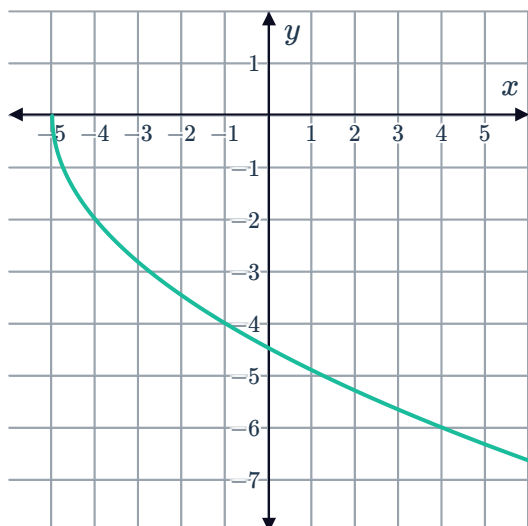
a

i

ii $x \geq -1$ iii $y \leq 0$

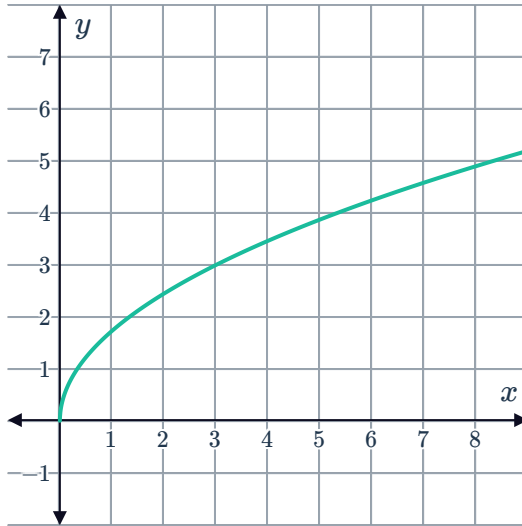
b

i

ii $x \geq -5$ iii $y \leq 0$

c

i

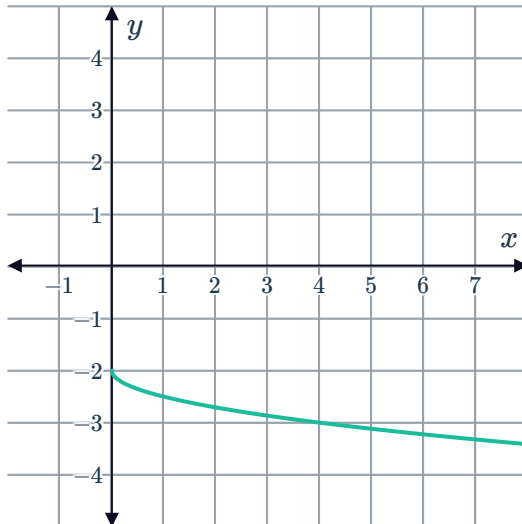


ii $x \geq 0$

iii $y \geq 0$

d

i

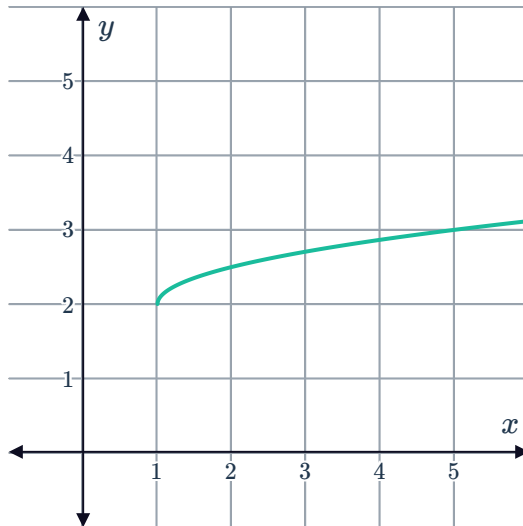


ii $x \geq 0$

iii $y \leq -2$

e

i



ii $x \geq 1$

iii $y \geq 2$

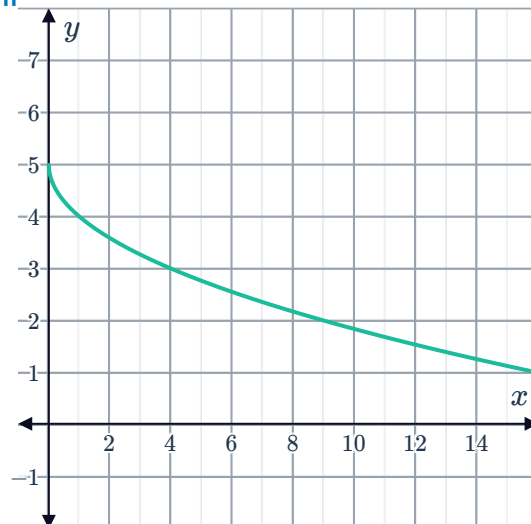
16

a

i $x \geq 0$

ii $y \leq 5$

iii

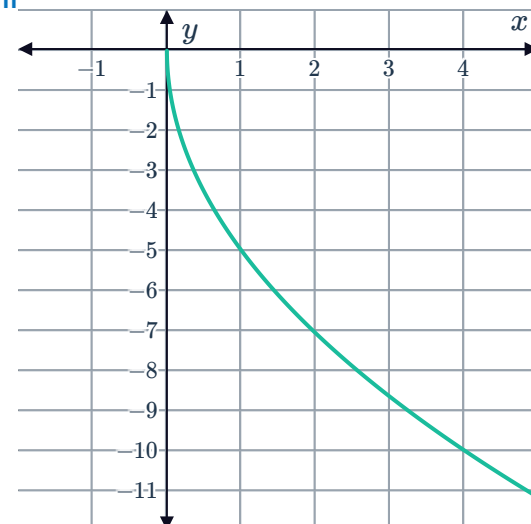


b

i $x \geq 0$

ii $y \leq 0$

iii



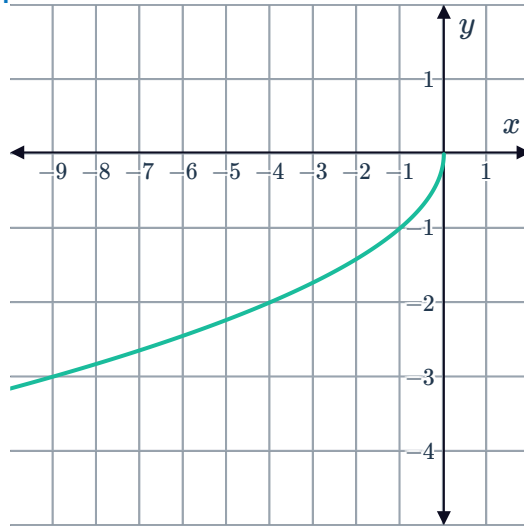
c

i $x \leq 0$

ii $y \leq 0$



iii

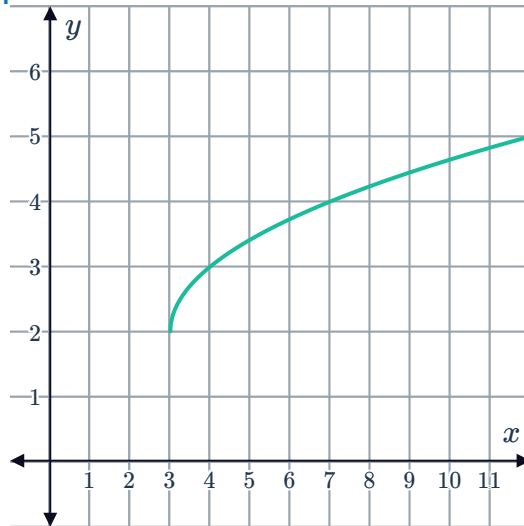


d

i $x \geq 3$

ii $y \geq 2$

iii



17

a $x \geq 0$

c $x \leq 3$

e All real values of x .

b $x \geq 2$

d $x \leq 3$

18 $[2, \infty)$

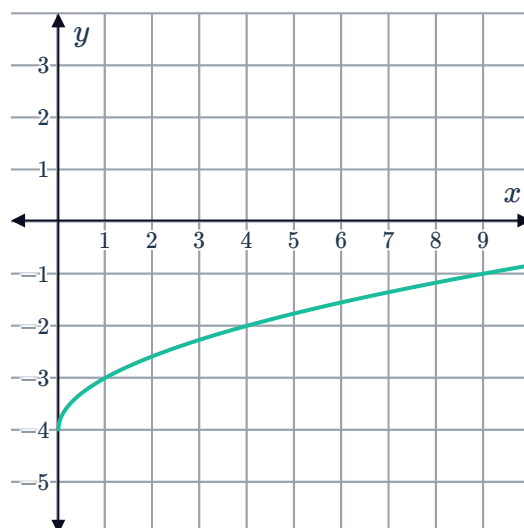
Transformation

19 $(0, 3)$

20

a The original function was translated down 4 units.

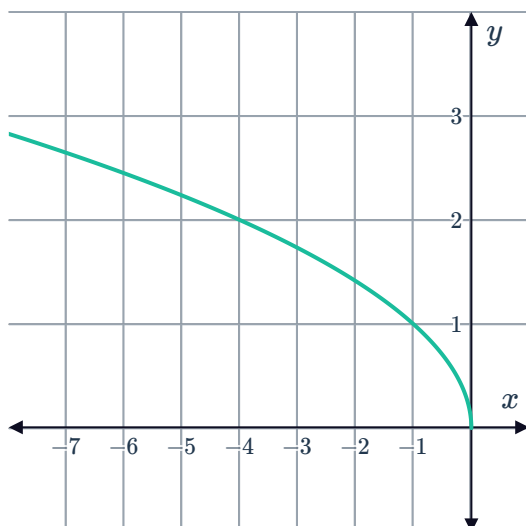
b



21

a

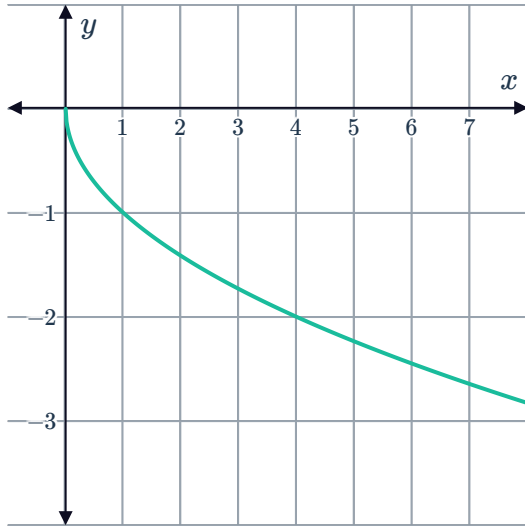
i



ii $y = \sqrt{-x}$

b

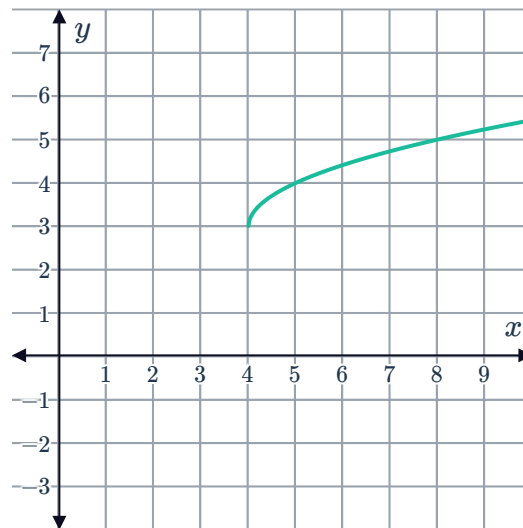
i

ii $y = -\sqrt{x}$

22

a Translate the graph 4 units to the right and 3 units up.

b



23

